PLTW POE

Activity 1.1.4 Powerful Pulleys

PLTW

POWERFUL PULLEYS

ACTIVITY SUPPORTS

Technology Transformation

GOALS

RESOURCES

- Design and validate a pulley system using CAD software.
- Investigate the factors that impact a pulley system.
- Understand the benefits and drawbacks of pulley systems.

GOALS

RESOURCES

- **Powerful Pulleys Onshape document**
- **Activity 1.1.4 Powerful Pulleys (Downloadable PDF)**

To appreciate the advancements in technology, it can be useful to examine tools and accomplishments of the past. Can you imagine the challenges of riding a horse over 2000 miles across the country on the Oregon Trail? Or how the Great Pyramid of Giza was built without modern tools? Nowadays, cars and trucks are essential to travel across the country, as well as computers and large cranes to help design and build massive structures.

Pulleys are another one of those essential machines that have changed how people move and lift objects. In this activity, you will explore and design pulley systems to move weight more efficiently.

Brute Force

To understand how useful pulleys are, look at how Fran has to load hay into her barn attic without a pulley as shown in Figure 1.

1. Review the Lifting Hay example to see how much force is required to lift the hay into Fran's loft.

Lifting Hay

Figure 1. Fran Lifting Hay

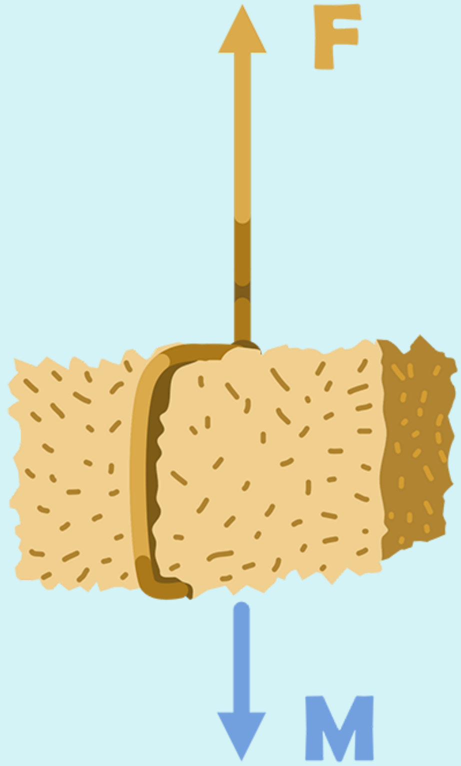
It's difficult lifting hay into a loft! Before pulleys, farmers may have simply used a rope, as shown in Figure 1. If Fran wanted to lift 45 kg (about 100 pounds) of hay into her barn loft, how much force would it require?

Using Newton's Second Law of motion, Force = Mass × Acceleration, and a free body diagram, you can calculate the amount of Force needed to pull the hay.

$F = m \cdot a$

$F = 45 \text{ kg} \cdot 9.81 \text{ m/s}^2$ (gravity)

$F = 441 \text{ N}$



Without a pulley, Fran would have to pull with a force of 441 N, or about 100 pounds, just to hold the hay in **static equilibrium**. Depending on how much hay she needs to store, that could be exhausting and time-consuming!

Powerful Pulleys

Fortunately for Fran, pulleys are a useful tool to help lift hay into the attic. You and your partner will use a simulated pulley tool to explore how pulleys can be used to increase mechanical advantage and decrease the amount of effort force needed to lift objects.

2

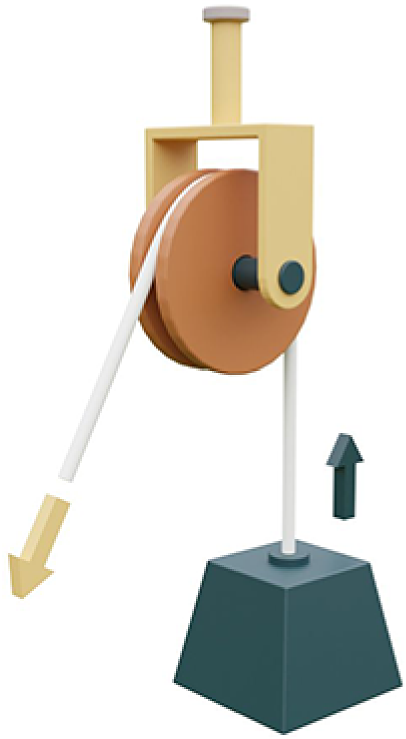
Review the Types of Pulleys presentation to see how they function.

Types of Pulleys

Learn more about...

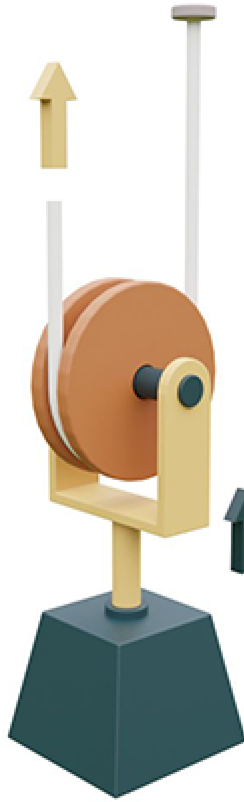
FIXED PULLEY	MOVABLE PULLEY	COMPOUND PULLEY

The pulley is attached to a fixed point, and the rope is attached to the weight that needs to be lifted.



FIXED PULLEY | **MOVABLE PULLEY** | COMPOUND PULLEY

The rope is attached to a fixed point, and when the free end of the rope is pulled, the pulley lifts the object.



FIXED PULLEY	MOVABLE PULLEY	COMPOUND PULLEY

This pulley is made of both fixed and movable pulleys. When the free end of the rope is pulled, the lower pulley lifts the object.

3

Navigate through the flashcards, and with a partner, identify which type of pulley each example shows.

movable pulley

compound pulley

fixed pulley

Turn and Talk

What do you think the ideal mechanical advantage is for the flag pole example? Why do you think this?

HISTORICAL HIGHLIGHT

Pulleys for Hay Stacking

Made of grass and other dried plants, hay is a common source of animal food known as *fodder*. Hay is used to feed cattle, horses, and other livestock on farms. Because hay is sensitive to weather, it is often stored in barn lofts or attics to keep it dry. Hay bales can weigh as little as 40 pounds and as much as 2000 pounds! In the late 1800s, farmers used a series of pulleys to move large amounts of hay (Figure 2).

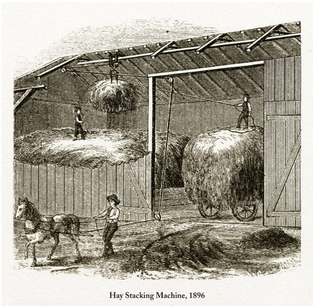


Figure 2. Hay Stacking Machine

Figure 3. Modern Conveyor Belt

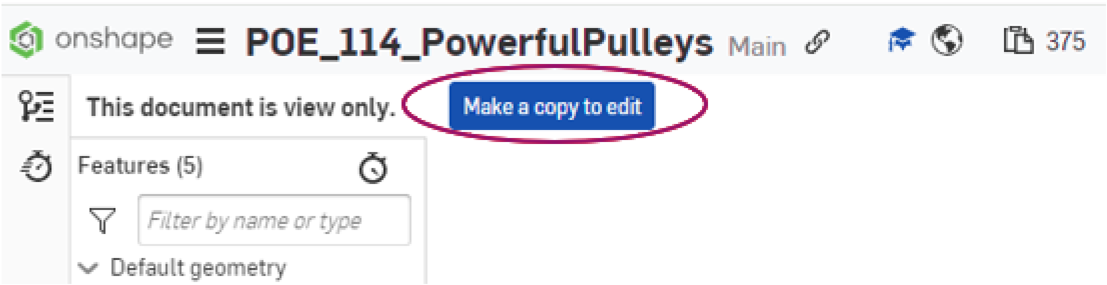
Nowadays, on larger farms, an inclined conveyor belt (Figure 3) is a common method of moving hay, which requires even less effort and time than a pulley system. However, pulleys are still widely used on small farms and in other applications where heavy objects must be moved or otherwise lifted.[1]

Exploring Pulleys

4

In Onshape®, create a copy of the **Powerful Pulleys Onshape document**. This Onshape document contains a model of a pulley that you can manipulate to calculate the maximum amount of weight lifted and the distance the weight is lifted in a designed pulley system.

- Select **Sign in** to open a project in OnShape.
- Select **Make a copy to edit** and name the document something unique.



5

On the right side of the interface, click the **Custom Tables** icon to open the Custom Table for the pulley system. The table contains all of the variables that are manipulated in the model as well as the Distance Lifted value.

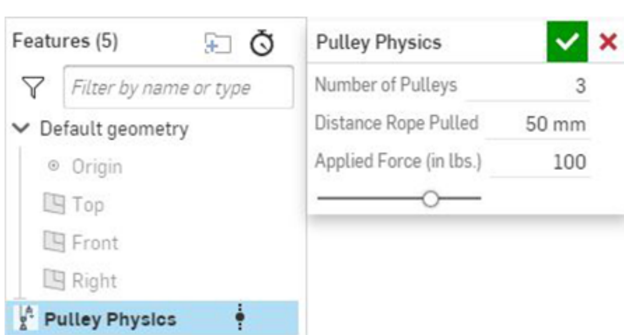
6 Design a pulley system for Fran that can lift a bale of hay that weighs at least 600 pounds with an applied force of less than 100 pounds.

Pulley Physics FeatureScript

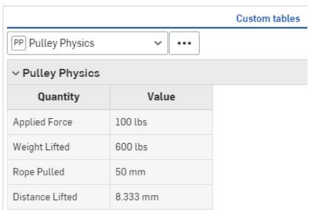
FeatureScript™ is a new programming language designed by Onshape for building and working with 3D parametric models. Using the FeatureScript language, users can create custom features within Onshape that extend the capabilities of the CAD platform. You will use a few custom features throughout this course.

This Pulley Physics FeatureScript is designed to allow you to manipulate the number of pulleys in the system, the distance a rope in the system can be pulled in millimeters, and the applied force to the system in pounds.

To change these variables, double-click **Pulley Physics** on the left side of the Onshape interface.



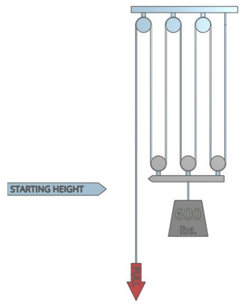
When you change a variable, the values change accordingly in the Pulley Physics Custom table on the right side of the interface.



Custom tables

Pulley Physics

Quantity	Value
Applied Force	100 lbs
Weight Lifted	600 lbs
Rope Pulled	50 mm
Distance Lifted	8.333 mm



The number of pulleys in the system, amount of weight lifted, and distance of rope pulled also change visually to match the variables.

Reflection

How much applied force does Fran have to exert to lift this mass now? Why do you think that is?

7 Now that you have designed a pulley system, continue to explore the variables in your pulley system to determine the relationship between the applied force, distance rope pulled, and number of pulleys, and how these affect the distance lifted and weight lifted.

- Continue to change the value of different variables to notice patterns within your pulley systems.

REMINDER: Remember to identify your independent and dependent variables to consistently track changes!

8 Use a spreadsheet tool or your PLTW Engineering Notebook to create plots of the data you collected and

describe the trends you notice.

Reflection

- Based on the data you collected, what is the most efficient way to lift a mass with a pulley system? Why?
- What affects the mechanical advantage of a pulley system? How do you know? Use examples from your data to support your answer.
- What is a tradeoff for having a mechanical advantage of 2 in a pulley system?

Application of Pulleys

In Activity 1.1.3 Mechanical System Efficiency, you explored an automated system that converted electrical energy into mechanical energy. Now, you will explore similar calculations with a pulley system.

9

In your engineering notebook, answer the following questions:

- Fran can lift 300 pounds of hay into her 15-foot loft in 10 minutes. How much power is required to complete this task?
- Instead of lifting the 300 pounds of hay into the 15-foot loft by herself, Fran decides to use a horse that can pull with 100 Watts of power (similar to Figure 4). How long will it take her now?


Hay Stacking Machine, 1896

Figure 4. Hay Stacking Machine

Conclusion

Question 1

Why are pulleys so useful? Provide evidence from your Onshape® designs.

Question 2

Investigate a modern application of pulleys. Explain how it is used and how its mechanical advantage improves the efficiency of the task being completed.

Source

Third-party images, videos, and Historical Highlight information used throughout this activity were drawn from the following sources:

1. PBS Online. (2003) Death of the Dream Farmhouses in the Heartland. [Alexandria, Va.?: PBS] [Web.] Retrieved from the Library of Congress, **https://lccn.loc.gov/2003618013**